Reliable Strictly Local Floorplan Editing under Imperfect Controls

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Abstract

Floorplan editors often report edit accuracy without showing whether unrelated geometry remains unchanged. We study a complementary requirement: reliable local execution under an explicit control interface. A request consists of an old CAD state, a declared edit region, and a typed control change. The output may change only inside the declared region and must remain structurally legal. We evaluate FloorFlow in two regimes. Under declared oracle controls, five operations reach Delta-IoU 0.9567–0.9998 and Delta-MSE 0.000019–0.000096, while preserving outside pixel, room, polygon, wall, opening, and edge content exactly. Under target-free predicted controls, the model produces structurally plausible plans, executes every supported validation prompt, and abstains from unsupported or ambiguous requests. We separate these safety properties from target-comparative fidelity: predicted Room macro IoU remains 0.2697 and operation-specific semantic success is mixed. The results position strict locality and safe operation execution as measurable requirements distinct from reconstruction accuracy.

1 Introduction

Local editing is more than generating a plausible replacement image. An edit must realize a requested operation while leaving unrelated architecture unchanged. Existing floorplan editors emphasize final layout quality or instruction following, but do not routinely expose a mechanical locality contract. This makes it difficult to distinguish a semantic edit from a global regeneration that happens to preserve the target.

We formulate editing as a control-state transition. The pre-edit drawing and its control state are valid inputs, as they are in an engineering workflow. A typed operation changes the control source; the model must realize that change in the current plan without modifying the outside region. Unsupported or ambiguous requests are not forced through the generator and instead produce a safe abstention.

Our contribution is a two-axis evaluation rather than a claim of complete-method SOTA:

- a strict locality and typed-operation contract;
- an oracle-control upper bound for edit fidelity;
- target-free structural reasonableness under imperfect controls;
- an operation-success audit that separates safe execution from requested semantic realization;
- matched local controls and deterministic sensitivity tests.

2 Related Work

HouseMind unifies floorplan understanding, generation, and editing with discrete room-instance tokens [1]. Its public edit table reports edit and graph metrics but does not expose strict outside preservation or attack-tested locality. Raster2Seq reconstructs vector plans from raster inputs and serves here as a matched reconstruction wrapper, not an editing-aware model [2]. CE2EPlan addresses boundary-conditioned vector generation [3]; generation evidence is not interpreted as editing evidence.

Floorplan datasets and metrics supply the geometric substrate. CubiCasa5K provides heterogeneous annotated floorplans [4]. GSDiff informs typed graph evaluation [5]. We report aligned, canonical-instance, and typed graph protocols separately and never pool their values.

3 Protocol

3.1 Control Transition

Let x_0 be the old visible plan and c_0 its predicted control state. A request consists of a normalized or control-grid edit region R and a typed operation a. The edited control is $c_1 = T_a(c_0)$. The model outputs $x_1 = f(x_0, c_1, R)$.

The old control source is part of the request. This mirrors CAD editing, where a new constraint is applied relative to an existing drawing and its prior controls. Successful operation behavior therefore requires two separate checks: whether the request is supported and whether the current plan realizes the requested change.

3.2 Strict Outside Domain

The valid outside domain is

$$O = P \wedge \neg R \wedge \neg D_1(W_0 \vee W_1), \tag{1}$$

where P is the footprint, W_0, W_1 are pre/post wall support, and D_1 is one-pixel dilation. Pixel, room-label, polygon, wall, opening, and unaffected-edge content outside R is compared with zero tolerance.

3.3 Typed Operations and Abstention

Supported operations are room-type change, room addition, room removal, opening deletion, and furniture deletion. Unsupported wall movement, ambiguous global scope, and conflicting instructions trigger abstention. Abstention returns the pre-edit plan and is never counted as a semantic success.

4 Method

FloorFlow composes a typed local delta with the cached pre-edit plan. Room operations modify the room partition; object deletions remove the requested family. The resulting candidate is repaired to a legal partition and rendered with explicit room, wall, opening, and furniture layers.

The operation audit derives the requested source and target room classes from the signed control-logit change. Room type and addition require the target class to account for at least half of changed pixels and contain at least four output pixels. Room removal requires the source class to be removed or merged. Object deletion requires the requested family to be absent after the local edit.

5 Experiments

5.1 Data and Evaluation

We use CubiCasa5K. Oracle-control editing uses test503 with five typed operations and five repeated seeds. Predicted-control plausibility uses val491 with five seeds and target-free generation. Predicted operation auditing uses validation outputs; the predicted model and operation thresholds are not frozen, so these numbers are preliminary. The test manifest used during development is not presented as a held-out result.

We report edit fidelity, exact outside preservation, FP-Eval validity, the independent FEVP five-check validity, raw and functional watertightness, graph diagnostics, structural plausibility, and operation-specific success. These metrics answer different questions and are not collapsed into one score.

5.2 Oracle-Control Fidelity

Table 1 reports the declared-control upper bound. All five operations numerically exceed the published HouseMind-E Delta-IoU and Delta-MSE anchors. The datasets and inputs differ, so this is protocol context rather than a leaderboard rank.

5.3 Predicted-Control Reliability

Predicted-control outputs are structurally plausible despite low target-comparative fidelity. Over val491, raster plausibility is 0.9849, plan plausibility 0.9866, functional watertightness 0.9980, and wall continuity 0.9954. Room macro IoU is only 0.2697. Raw fpeval watertightness is 0.8167; the gap is concentrated in multi-component footprints, while all 401 single-component footprints reach raw watertightness 1.0.

The latest development exposure contains 738 prompts: all 438 supported prompts execute, all 300 unsupported or ambiguous prompts abstain, and every category preserves outside content exactly with validity 1.0. Because these rows used the test manifest during development, they are not held-out evidence.

5.4 Operation Realization

Table 2 reports preliminary validation semantic success. Safe execution alone is insufficient: room addition reaches 0.92, room type 0.55, room removal 0.20, opening deletion 0.51, and furniture deletion 0.00. These rates are retained as a boundary and will be recomputed after model freeze.

5.5 Mechanism and Robustness

Typed routing improves aggregate Delta-IoU over dense, no-gate, and no-graph arms while retaining exact outside room preservation. At matched observable compute with no added learned parameters, typed routing raises Delta-IoU by 0.0487 and lowers GED by 0.0614. The strict arm has five-layer outside exactness 1.0 and violation rate 0.0 over 12,395 rows and 7,345 single-room controls. Twenty-six deterministic attacks are all sensitive with zero observed false passes or false failures.

6 Limitations

Oracle controls are target-derived and non-deployable. Predicted reconstruction and graph fidelity remain weak. Semantic operation success is mixed, and the current development run is not held-out evidence. The model, thresholds, and evaluator must be frozen before a fresh one-shot evaluation. The paper does not claim complete HouseMind superiority, RPLAN SOTA, full-CAD SOTA, or free-text editing.

7 Conclusion

Strict locality and safe operation execution are measurable properties distinct from reconstruction fidelity. FloorFlow provides a strong oracle-control editing anchor and a plausible target-free execution path, while the preliminary operation audit exposes exactly where semantic realization remains unsolved.

References

[1] HouseMind: Unified Floorplan Understanding, Generation, and Editing. arXiv:2603.11640.

[2] Raster2Seq: Vector Floorplan Reconstruction from Raster Inputs. DOI:10.1145/3799902.3811124.

[3] CE2EPlan: End-to-End Boundary-Conditioned Vector Floorplan Generation. IEEE TVCG, 2026. DOI:10.1109/TVCG.2026.3665422.

[4] CubiCasa5K: A Dataset and Benchmark for Floorplan Recognition. DOI:10.1007/978-3-030-20205-7_3.

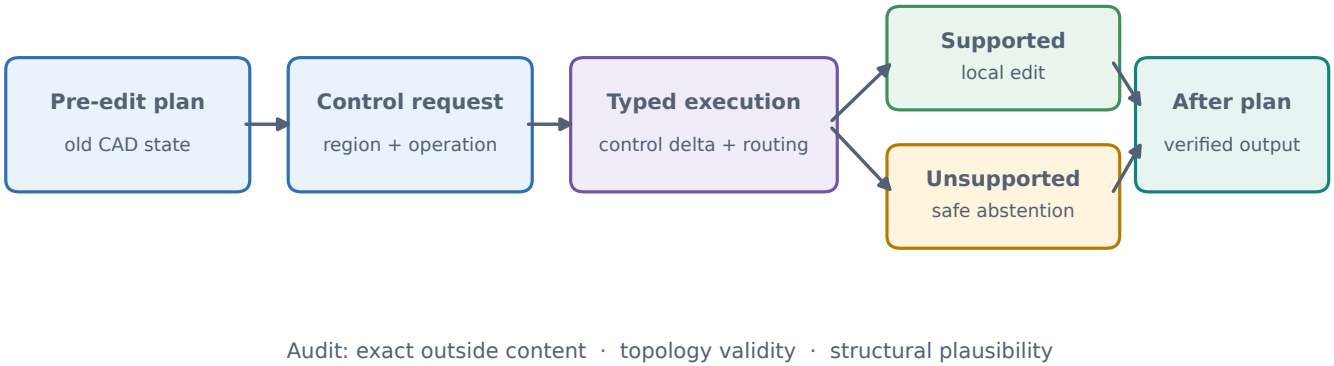


Figure 1: The editing contract. The old CAD state and typed control request enter the executor. Supported requests receive a local edit, while unsupported or ambiguous requests abstain. Accepted outputs are checked for exact outside preservation, topology validity, and structural plausibility.

Table 1: Oracle-control editing on CubiCasa test503. Outside exact covers pixel, room, polygon, wall, opening, and unaffected-edge content.

Operation	N	Delta-IoU	Delta-MSE	Outside exact	FP-Eval
Room type	487	0.999199	0.000069	1.000000	1.000000
Room add	495	0.999801	0.000030	1.000000	1.000000
Room remove	500	0.999756	0.000019	1.000000	1.000000
Opening delete	502	0.956703	0.000096	1.000000	1.000000
Furniture delete	495	0.987812	0.000083	1.000000	1.000000

Table 2: Preliminary semantic operation success. Denominator is the number of supported prompts. Safety and semantic success are separate claims.

Operation	Success	Denominator
Room add	0.92	100
Room type	0.55	100
Room remove	0.20	100
Opening delete	0.5085	59
Furniture delete	0.00	15

[5] GSDiff: Geometry-Enhanced Structural Graph Generation. DOI:10.1609/aaai.v39i16.33904.

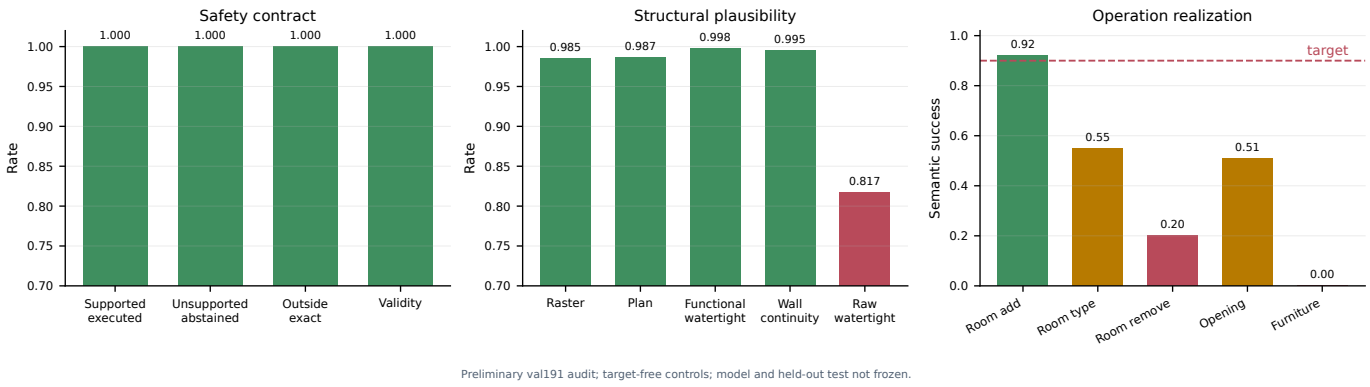


Figure 2: Safety, structural plausibility, and semantic operation realization are separate axes. The current model satisfies the safety contract and produces plausible plans, while semantic realization remains mixed. This is a preliminary validation audit, not a held-out result.